

SQL

Ch-1 (Why Database)

File System vs DBMS

A file system (managed by OS) is responsible for creating/deleting files or reading/writing bytes or managing storage on disk.

Before database, most software stored info directly in OS files.

But the OS manages files not the data semantics. Concepts like tables, relationships, keys, constraints and transactions are implemented by DBMS, not file system.

Flat files like .csv .txt can store data but they cant efficiently manage data at scale.

What is a DBMS?

A database management system (DBMS) is a software that provides an abstraction layer between application and physical storage. Emphasizes management not just storage.

Provides :-

I)

Persistent storage

data survives after program exit

II)

Efficient retrieval

instead of scanning every record, DBMS uses indexes and access paths to retrieve data quickly

III)

Data integrity

constraints ensure invalid data cant be stored

IV)

Concurrency control

thousands of users can read and modify data at same time without corrupting it

V)

Transaction management

a group of related operation either all succeed or fail together

VI)

Crash recovery

if system crashes after transaction committed, DBMS can recover committed data and roll back incomplete work

VII)

Security

different users have different info

VIII)

Data independence

interact with logical view of data without worry on how it's stored on disk

Scaling & File-Based Systems

For simple sequential storage flat files can be faster as less overhead, but as data volume, complexity, concurrency increases DBMS performs better due to indexing, buffering, caching, query optimization.

File Based System become difficult to maintain as app grows ↑, multiple programs have their own copies of data and implement their own logic leading to :-

- Data redundancy
- Data inconsistency
- lack of concurrency control
- poor scalability
- complex recovery mechanism

DBMS gives ↓

centralized organization of data, efficient querying, indexing, query optimization, data integrity via constraints, transaction management, concurrency control, crash recovery, security, data independence, recovery mechanisms